CST515 - Computer Security

CST 515 (Computer Security) Syllabus

Course Information

Course Details

- **Course Title:** Computer Security
- **Course Number:** CST 515
- **Meeting Location:** Physical
- **Units:** 4

Course Description

This course covers key concepts in computer security, including cryptography, access control, operating systems security, network security, software security, and blockchain technology. Students will also explore malware analysis, social engineering, and strategies to address modern cybersecurity challenges by understanding both technical and human factors in security.

Materials

- Course textbook: "Computer Security: Principles and Practice" by William Stallings and Lawrie Brown

Technology Requirements

Students must have access to: - A computer (not "chromebook" or similar) with reliable internet connection - Access to Canvas learning management system

Course Objectives

At the end of this class, you should be able to:

1. Apply cryptographic principles and implement secure access control mechanisms
2. Analyze and mitigate security vulnerabilities in operating systems and network environments
3. Understand blockchain technology and its security implications
4. Identify and respond to malware threats and social engineering attacks

Grading

There are three groups of assignments, briefly described in the table below. **Getting below a 40% in any of these three groups will result in failing the class.**

Assignment Kind	Value
Projects	35%
Exams	35%
Participation	30%

Assignment Kind	Value

Projects

Projects will take the form of penetration testing labs, security audits of real systems, cryptographic implementation assignments, and vulnerability assessment exercises. Students will also conduct research on emerging cybersecurity threats and develop defensive strategies.

Exams

Exams will cover material from class and from any assigned reading, as well as draw from knowledge gained during projects.

Participation

Class is expected to be highly interactive, with students actively answering and asking questions, as well as participating in discussion groups. Therefore, attendance is required, as well as participation in all in-class activities. These activities may take the form of in-class discussions, brainstorming sessions, stand-ups, as well as making active progress on projects during class time.

Grade Assignment

Grades as assigned based on the standard ranges, which are outlined below.

Grade Range	Letter Grade
$[97, \infty)$	A+
$[93, 97)$	A
$[90, 93)$	A-
$[87, 90)$	B+
$[83, 87)$	B
$[80, 83)$	B-
$[77, 80)$	C+
$[73, 77)$	C
$[70, 73)$	C-
$[60, 70)$	D
$[0, 60)$	F

Syllabus Change Policy

This syllabus is subject to change at the instructor's discretion to enhance learning or accommodate unforeseen circumstances. Students will be notified of any substantive changes (such as changes to due dates, point values of assignments, or major policy modifications) via Canvas announcement and/or email at least one week in advance when possible.

The process for syllabus changes: 1. Instructor identifies need for change 2. Revised syllabus section is prepared 3. Students are notified via Canvas and email 4. Change takes effect after notification period

Students are responsible for staying current with any announced changes to the syllabus.